

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1598

Alchem 138 kV
May 28, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1598. The Interconnection Customer proposes a 221.5 MW wind facility interconnecting at the Alchem 138 kV substation in the New Jersey area. The request was received on October 18, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 474 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,153,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
S Tacoma – Alchem 138 kV	Dam Hights – Babine Lake 138 kV	110.7	315	31.2
Maywood – Pacific Power 138 kV Line	S Tacoma – Alchem 138 kV	107.1	236	26.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
S Tacoma – Alchem 138 kV	Rebuild line with higher-capacity conductor	\$2,803,000	64% (\$1,794,000)
Pacific Power – Maywood 138 kV Line	Reconductor line	\$534,000	67% (\$359,000)
	Total allocated to project		\$2,153,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.